

Simulation and Modeling

Lecture 08

Generating Random Variates

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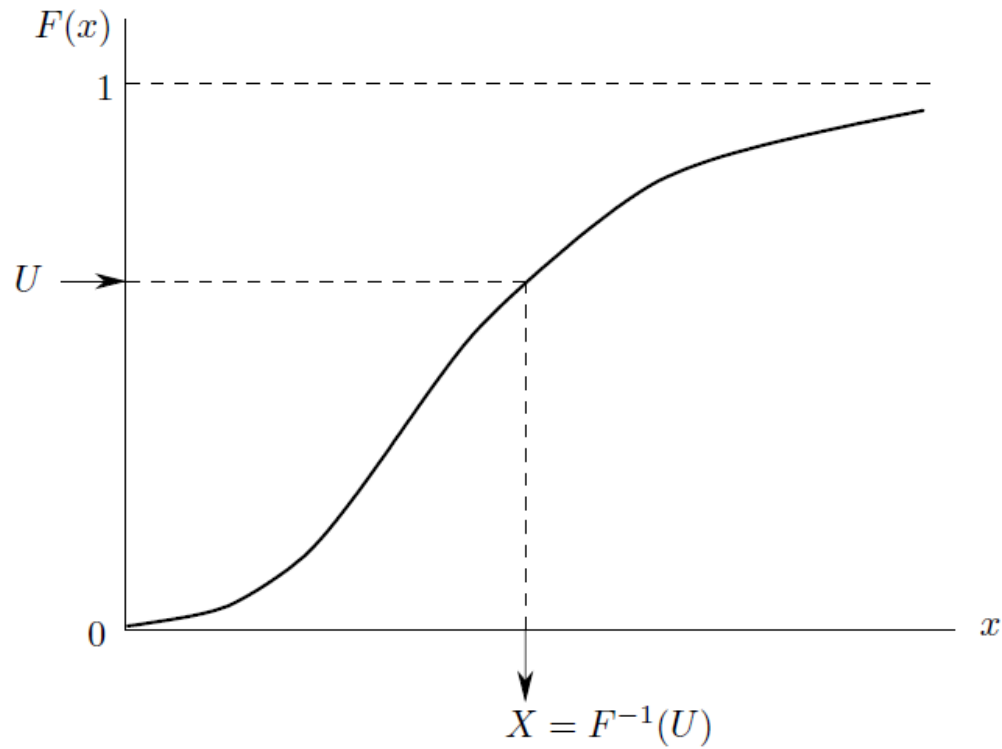
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Introduction

- Generating Random Numbers
 - Obtaining sample values of a random variable having uniform continuous distribution in $[0, 1]$
- Generating Random variates
 - Obtaining sample values of a random variable having the desired distribution

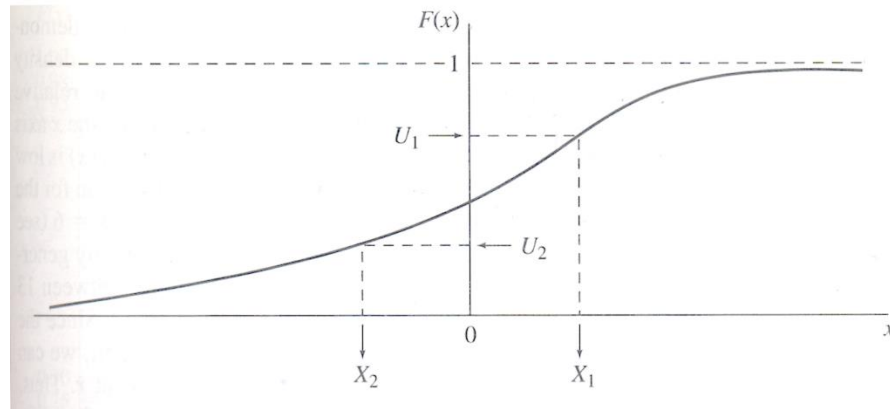
Inverse Transform Method



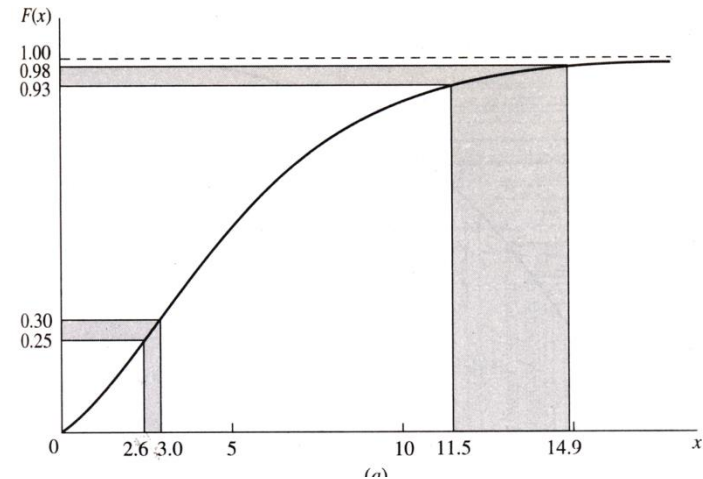
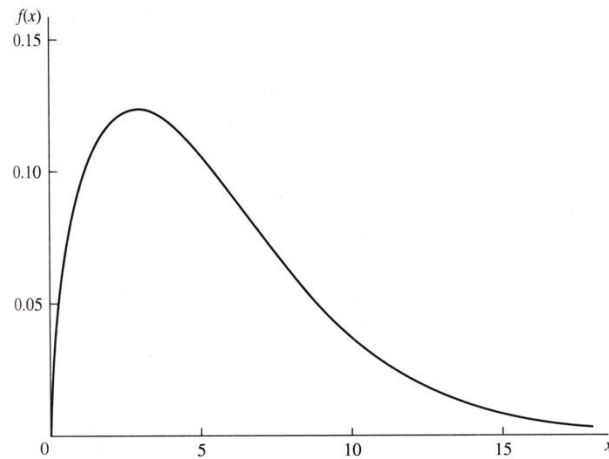
Inverse Transform Method

- The PDF of a r.v. X , $f_X(x)$
 - Relative chance of observing a value of X on different part of its range
 - If $f_X(x)$ is high for a region
 - We observe a lot of values from that region
 - $f_X(x)$ is derivative of $F_X(x)$ and represents the slope of $F_X(x)$ at X
 - $F_X(x)$ is steep , i.e., increases at a high rate, when $f_X(x)$ is high
 - $F_X(x)$ flat when $f_X(x)$ is small

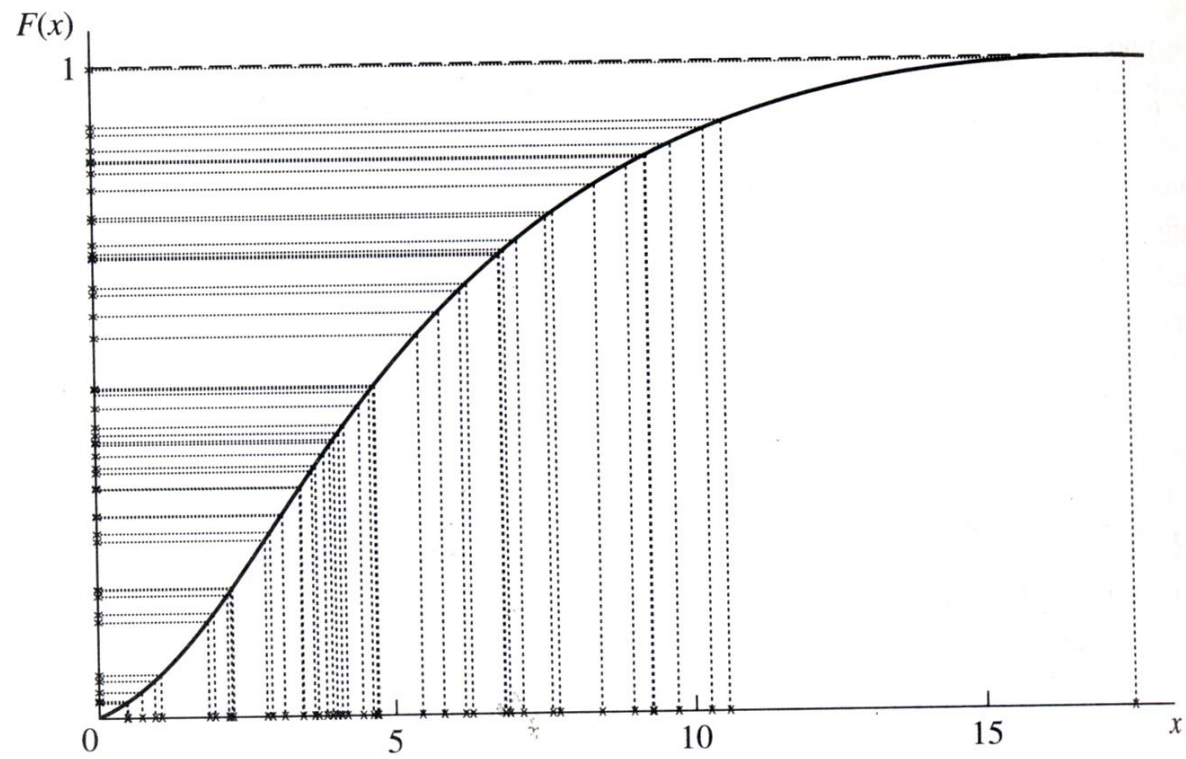
Inverse Transform Method



- U is uniformly distributed on F_X
- More U s hit the steep part of F_X than the flat part
- More values of X will be observed from the steep part than the flat part.



- For the interval $[0.25 - 0.30]$ (5% U 's)
 - Corresponding X values will be in $[2.6 - 3.0]$
 - A narrow region
- For the interval $[0.93 - 0.98]$ (5% U 's)
 - Corresponding X values will be in $[11.5 - 14.9]$
 - A larger interval



Inverse Transform Method

- Let U be a uniform $(0, 1)$, $U \sim U(0, 1)$ RV

- F is any continuous CDF, and
- X is a random variable defined by

$$X = F^{-1}(U)$$

- X has the distribution F , i.e., $F_X(x) = F$
 - $F^{-1}(U)$ is defined to be that value of X , so that
$$U = F_X(x)$$

- Proof: Let F_X denote the CDF of X

$$F_X(x) = P[X \leq x] = P[F^{-1}(U) \leq x]$$

- If $a \leq b$, then $F_X(b) \leq F_X(a)$
- $F_X(x) = P[F(F^{-1}(U)) \leq F(x)] = P[U \leq F(x)]$
- Since U is uniform $[0,1]$, $F_X(x) = F(x)$

Inverse Transform Method: Algorithm

- Step 1: Generate $U \sim U(0,1)$
- If not available, derive the CDF of the distribution
- Set $X = F^{-1}(U)$
- Return X

Example 1

- Generate random variates for the following distribution

$$F(x) = x^n$$

- Let $x = F^{-1}(u)$, then
- $u = F(x) = x^n$
- $x = u^{\frac{1}{n}}$

Example 1: Algorithm

- Step 1: Generate $u \sim U(0, 1)$
- Step 2: Calculate $x = u^{\frac{1}{n}}$
- Step 3: Return x

Example 2: Exponential Distribution

- Generate random variates with exponential distribution having rate λ and pdf

$$f_X(x) = \lambda e^{-\lambda x}, \quad 0 \leq x \leq \infty$$

$$F_X(x) = \int_0^x \lambda e^{-\lambda x} dx = 1 - e^{-\lambda x}$$

- Let $x = F_X^{-1}(u) \Rightarrow u = F_X(x) = 1 - e^{-\lambda x}$
- $e^{-\lambda x} = 1 - u$
- $-\lambda x \ln e = \ln(1 - u)$
- $x = -\frac{1}{\lambda} \ln(1 - u)$

Example 2: Exponential Distribution

- Algorithm
 - Step 1: Generate $u_i \sim U(0, 1)$
 - Step 2: Calculate $x_i = -\frac{1}{\lambda} \ln(1 - u_i)$
 - Step 3: Return x_i
- $(1 - U)$ is also uniform in the range $(0, 1)$
- $-\frac{1}{\lambda} \ln(1 - U)$ has the same distribution as $-\frac{1}{\lambda} \ln(U)$
 - A small savings in time can be achieved

Example 3: Uniform Distribution

- $f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$
- Step 1: calculate the CDF
 - $F_X(x) = \int_a^b \frac{1}{b-a} dx = \frac{x-a}{b-a}$
- Step 2: Let $x = F_X^{-1}(u) \Rightarrow u = F_X(x)$
 - $\Rightarrow u = \frac{x-a}{b-a} \Rightarrow x - a = u(b - a)$
 - $x = a + u(b - a)$

Example 3: Uniform Distribution

- Algorithm
 - Step 1: Generate $u_i \sim U(0, 1)$
 - Step 2: Calculate $x_i = a + u_i(b - a)$
 - Step 3: Return x_i

Example 4: Weibull Distribution

- $f_X(x) = \begin{cases} \frac{\beta}{\alpha^\beta} x^{\beta-1} e^{-\left(\frac{x}{\alpha}\right)^\beta}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$
- Step 1: $F_X(x) = 1 - e^{-\left(\frac{x}{\alpha}\right)^\beta}$
- Step 2: $x = F_X^{-1}(u) \gg u = F_X(x) = 1 - e^{-\left(\frac{x}{\alpha}\right)^\beta}$
- Step 3: derive the solution of x
$$x = \alpha[-\ln(1 - u)]^{1/\beta}$$
- Step 4: Return x

Example 4: Weibull Distribution

- Algorithm
 - Step 1: Generate $u_i \sim U(0, 1)$
 - Step 2: Calculate $x_i = \alpha[-\ln(1 - u_i)]^{\frac{1}{\beta}}$
 - Return x_i